

Lucas Leong

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EXPERIENCE

Software Engineer Intern

Aug. 2025 – Oct. 2025

Precix

precix.io

- Explored reverse engineering of proprietary ultrasound firmware DLLs and OpenCL kernels to study real-time medical imaging pipelines using IDA Pro and windbg
- Developed embedded C firmware for IMUs and ultrasound probes, implementing multithreaded data acquisition and wireless transmission on Raspberry Pi CM5 and ESP32-P4/C6.
- Prototyped end-to-end system integrating sensor and imaging data into an ElysiaJS backend with a Svelte-based frontend for real-time visualization using websockets.

Software Developer

Feb. 2024 – May. 2024

eas-ai.com

- Developed a full-stack web application using NextJS, Prisma, PostgreSQL and Docker, providing a platform for funeral parlors to sell funeral packages for customers

PROJECTS

Serotonin | Objective-C, C, iOS Kernel Exploitation

Jan. 2024

- Independently researched and implemented a novel iOS 16.0–16.6.1 semi-jailbreak; published open source, gaining **650+ GitHub stars** and **65,000+ downloads**
- Developed original kernel namecache vnode redirection technique to hijack launchd execution without breaking SSV, bypassing iOS kernel memory protections
- Reverse engineered iOS kernel internals (vnodes, PAC, launch constraints, codesigning) using IDA Pro and kernel read/write primitives from kfd exploit chain

Wayside – Hack&Roll 2026 Winner | C, ESP32, ESP-NOW, BLE, Embedded Systems

Jan. 2026

- **Won top 9**; built proximity-based networking hardware badges inspired by Nintendo StreetPass for passive discovery of compatible hackathon participants
- Implemented custom network stack from scratch using ESP-NOW for peer broadcasting and RSSI-based proximity estimation on ESP32 microcontrollers
- Integrated BLE pairing with personal devices and cryptographic layer for secure peer-to-peer data exchange between badges

Balatro Neural Network | Python, Lua, Deep Q-Learning, Smods

Feb.–May 2025

- Built a game mod using Smods and Lovely Injector exposing internal game state to a Python server over a custom client-server protocol
- Trained a Deep Q-Learning agent to make real-time in-game decisions; agent learned basic card selection strategies after training

BrainHack – TIL-AI | Python, Docker, TensorFlow

June 2023

- Achieved **8th place** in semifinals; finetuned vision models for zero-shot object detection deployed via Docker Compose onto physical robot hardware

SKILLS

Languages: C, Python, JavaScript, TypeScript, Bash, Lua, Kotlin, Rust

Systems & Security: IDA Pro, pwntools, gdb, WinDbg, ESP-IDF, Linux, Docker

Web: React, NextJS, Node.js, Bun, Elysia, Flask, FastAPI

Tools: Git, Vercel, Figma

EDUCATION

Nanyang Polytechnic

Diploma in Information Technology

Singapore, Singapore

Apr. 2023 – Apr. 2026

Singapore University of Technology and Design (SUTD)

Computer Science and Design

Singapore, Singapore

2026 – 2029